

PEAT REWETTING PLANNING SYSTEM

Domain-Driven Design Document

Business language, bounded contexts, aggregates, domain events and business rules of the peat rewetting planning domain.

Project	Peat Rewetting Planning for 337 Peat Hydrological Units (KHG)
Document	Domain-Driven Design
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Contents

Purpose and scope

Domain overview

Ubiquitous language

Subdomains

Bounded contexts

Context map

Aggregates

Entities and value objects

Plan Version lifecycle

Domain events

Domain services

Business rules catalogue

Mapping to the implementation

Open questions for domain experts

Purpose and scope

This document describes the business domain behind the Peat Rewetting Planning system using Domain-Driven Design (DDD). It explains the language used by the people who plan peatland rewetting, splits the problem into bounded contexts, and names the aggregates, rules and events that the database enforces.

It is written for three readers:

- **Presenters and project managers** who need to explain what the system models and why.
- **Engineers** who extend the database, the OGC API or a future web application.
- **Domain experts** who want to check that the model matches how planning and review actually work.

The implementation details (tables, columns, deployment) are covered in the companion documents: *Entity Relationship Diagram*, *Database Schema* and *System Architecture Document*.

Domain overview

Indonesia manages its peatlands in **Peat Hydrological Units** (Kesatuan Hidrologis Gambut, KHG). A KHG is a peat dome bounded by rivers, and it behaves as one hydrological system. Drained peat dries out and burns, so the national programme rewets it by:

1. **Blocking canals** with canal blocks (sekat kanal) placed where a canal crosses a contour line, so that each block holds back a fixed drop in water level.
2. **Pumping water** from rivers into the canal network during the dry season.
3. **Monitoring** groundwater level (TMAT), deep wells and fire hotspots.

Planning happens **per KHG**. An analyst prepares a plan in QGIS or in the web application, a reviewer from the ministry checks it against automatic validation rules, and an approved plan is printed as an A0 poster for field teams. Every change must be traceable to a person, a tool and a plan version.

The system currently holds 866 KHG, of which 345 contain the 2,004 target villages for 2026. The user interface calls this programme "337 KHG".

Ubiquitous language

The team uses Indonesian terms. The database keeps those names so that analysts recognise them; this table is the shared dictionary.

Term (database)	English term	Meaning	Stored in
KHG	Peat Hydrological Unit	Planning unit. Every plan, review and poster belongs to one KHG.	<code>ref.khg</code>
Sekat kanal	Canal block	Small dam in a canal. <i>Rencana</i> means planned, <i>existing</i> means built.	<code>peta.sekat</code>
Kanal	Canal	Drainage channel; primer, sekunder or tersier (primary, secondary, tertiary).	<code>peta.kanal</code>
Kanal aktif	Active canal	Canal that still carries water, confirmed by field survey.	<code>peta.kanal.status</code>
Pompa	Pump	Planned or installed pump that moves river water into canals.	<code>peta.pompa</code>
Pintu air	Water gate	Gate structure that controls canal flow.	<code>peta.pintu_air</code>
Sungai	River	Water source for pumps.	<code>peta.sungai</code>
Sumur bor	Deep well	Well used for fire fighting and rewetting, with daily discharge.	<code>peta.sumur_bor</code>
Logger AP-TMAT	Groundwater logger	Sensor that records groundwater depth; alert above 0.4 m.	<code>peta.logger_tmat</code>
Hotspot	Fire hotspot	Satellite fire detection (MODIS, VIIRS).	<code>peta.hotspot</code>
Areal terbakar	Burned area	Burn scar polygon with the list of years it burned.	<code>peta.areal_terbakar</code>
Ketebalan gambut	Peat thickness	Depth class of the peat layer. Peat at least 3 m deep has a protection function.	<code>peta.ketebalan_gambut</code>
Konsesi	Concession	Plantation or forestry permit area owned by a company.	<code>peta.konsesi</code>

Term (database)	English term	Meaning	Stored in
Unit analisis	Analysis unit	Polygon from overlaying KHG, village, concession, forest status, peat and burn history.	<code>analisis.unit_*</code>
Versi KHG	Plan version	Numbered snapshot of the plan for one KHG (v1, v2 and so on).	<code>alur.khg_versi</code>
Status alur	Workflow status	Draft, review, revisi (revision requested), approved, printed.	<code>ref.status_alur</code>
Perubahan	Change record	Audit entry for one create, update or delete.	<code>alur.perubahan</code>
Validasi otomatis	Automatic validation	Rule check run before review.	<code>alur.validasi_khg()</code>
Saran lokasi	Location suggestion	Ranked candidate sites for a new pump.	<code>peta.saran_lokasi_pompa()</code>
Ekspor poster	Poster export	Job that renders a plan as an A0 to A3 map.	<code>alur.ekspor_poster</code>
Via	Channel	Tool that made a change: qgis, web, api, sync or import.	meta column <code>updated_via</code>

Subdomains

Type	Subdomain	Why it has this type
Core	Rewetting infrastructure planning	This is where the programme creates value: where to put blocks and pumps.
Core	Plan review and versioning	Accountability to the ministry depends on traceable, approved plans.
Supporting	Fire and peat risk	Inputs that prioritise KHG and score pump locations.
Supporting	Field monitoring	Feedback from the ground (groundwater, wells) after construction.
Supporting	Spatial analysis	Pre-computed overlays that give every location its context.
Supporting	Publication	Posters and the OGC API that share results with other teams.
Generic	Reference data	Administrative regions, companies and code lists.
Generic	Identity and access	Database roles and HTTP Basic authentication.

Bounded contexts

Each bounded context owns a clear part of the model and speaks its own slice of the ubiquitous language. In this system a context maps to a group of tables inside one PostgreSQL schema.

Bounded context	Responsibility	Main objects	Schema
Reference	Regions, KHG boundaries, companies and code lists that every other context uses.	Province, Regency, District, Village, KHG, Company, lookups	ref
Infrastructure Planning	Canals, canal blocks, pumps, water gates and rivers of a KHG plan.	Canal, Canal Block, Pump, Water Gate, River	peta
Risk	Hotspots, burned areas, peat thickness and concessions.	Hotspot, Burned Area, Peat Thickness, Concession	peta
Field Monitoring	Groundwater loggers, deep wells and fire posts with their time series.	Logger, Reading, Deep Well, Discharge, Fire Post	peta
Spatial Analysis	National and target-area analysis units and contour lines. Read only.	Analysis Unit, Contour	analisis
Plan Review	Versions, change records, comments, validation, poster exports.	Plan Version, Change Record, Comment, Poster Export	alur
Publication	Stable public read model and OGC API.	Public views, OGC collections and processes	api

Context map

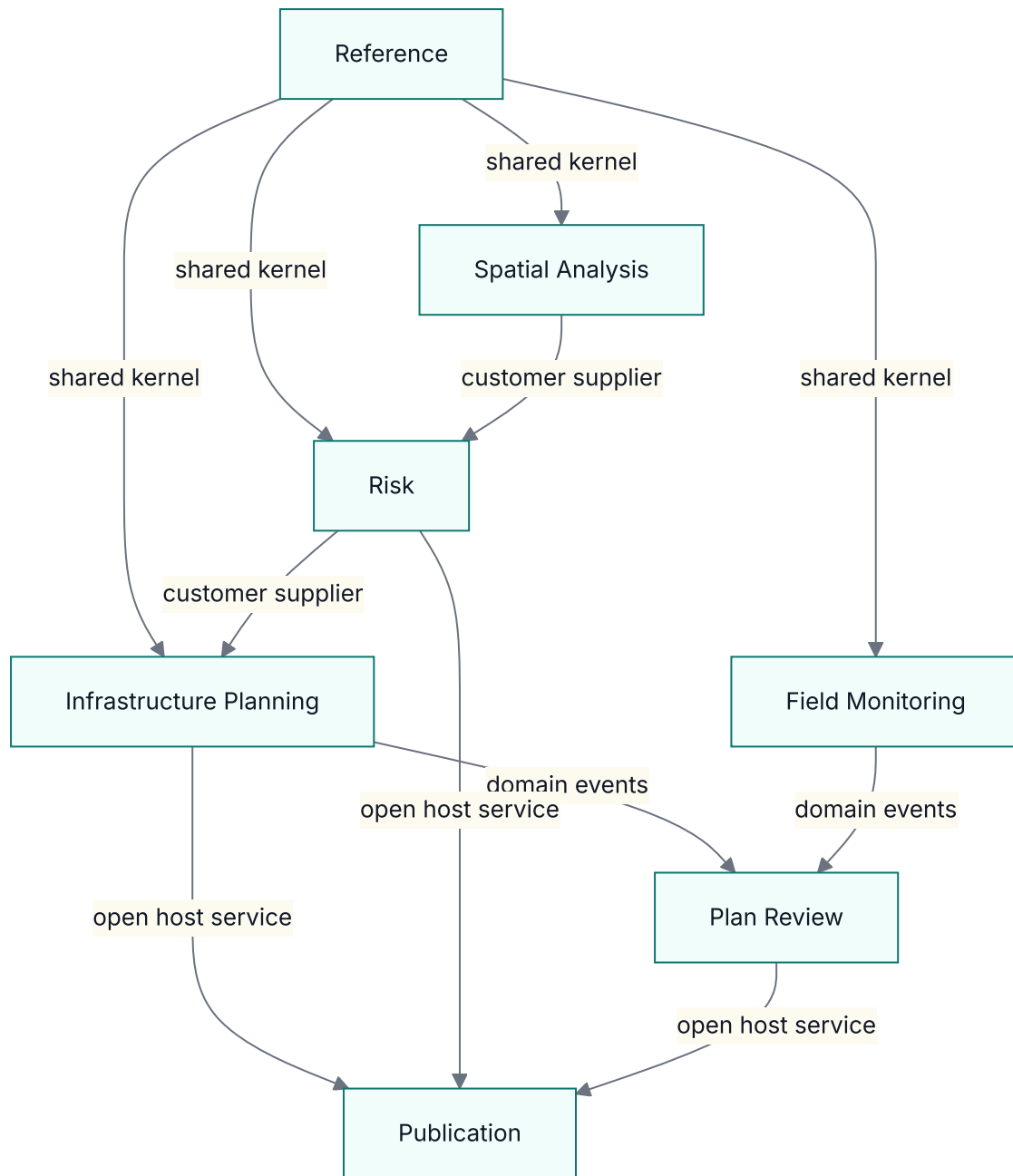


Figure 1. Context map between the internal bounded contexts.

External sources enter through an anti-corruption layer (ACL): the ETL scripts in `scripts/` and `db/04` to `db/08`.

External source	Enters context	What the ACL translates
BRGM PPEG canal blocks and hydrological infrastructure	Infrastructure Planning	Removes duplicates, fixes a wrong contour field, unifies codes, links blocks to canals by location
BlueBook PEG 2026 geodatabase	Reference, Spatial Analysis	Normalises names and units, dissolves KHG boundaries, concessions and peat thickness

External source	Enters context	What the ACL translates
FIRMS and SiPongi hotspot exports	Risk	Merges two file formats, converts local time zones to UTC, unifies confidence levels
Burn scar mapping 2015 to 2026	Risk	Converts twelve year columns into one list of burn years
Deep Well system (future)	Field Monitoring	Planned synchronisation keyed by external id

How to read the relationships:

- **Anti-corruption layer (ACL).** External shapefiles and geodatabases use inconsistent names, mixed units and duplicate rows. The ETL scripts (`scripts/` , `db/04` to `db/08`) translate them into the clean model, so no external format leaks into the core.
- **Shared kernel.** The Reference context (KHG, villages, companies, code lists) is shared by every other context and changes rarely.
- **Customer and supplier.** Spatial Analysis supplies the KHG boundaries, concessions and peat thickness that Risk uses. Risk supplies scores that Planning uses to rank pump locations.
- **Domain events.** Planning and Monitoring do not call Plan Review. Instead, every change emits a change record that Plan Review attaches to the open version.
- **Open host service.** Publication exposes a stable, standards-based interface (OGC API) so that outside teams do not depend on internal tables.

Aggregates

An aggregate is a cluster of objects that must stay consistent together. Changes go through the aggregate root, and invariants are enforced at that boundary. The table below lists the aggregates and where each invariant is enforced.

Aggregate (root)	Members	Key invariants	Enforced by
Plan Version (<code>khg_versi</code>)	Change records, comments, validation snapshot, poster exports	At most one open version (draft, review or revisi) per KHG. Version numbers increase by one. Status follows the allowed transitions.	Partial unique index, <code>alur.versi_terbuka()</code> , <code>alur.ajukan_review()</code> , <code>alur.putuskan()</code>
Canal Block (<code>sekat</code>)	Overlay context (value object), building details (entity)	Code <code>SK-n</code> unique within a KHG. Status from the code list. Point geometry in EPSG:4326. KHG derived from location when missing.	Unique index, FK, <code>isi_meta</code> , <code>isi_kode</code> triggers
Canal (<code>kanaḷ</code>)	None	Code <code>K-n</code> unique within a KHG. Length always equals the geodesic length of the geometry. A missing status means "not surveyed".	Unique index, generated column
Pump (<code>pompa</code>)	Suitability scores (value object)	Code <code>P-n</code> unique within a KHG. Each score between 0 and 1. Should be within 500 m of a water source and outside concessions.	CHECK constraints, validation function
Deep Well (<code>sumur_bor</code>)	Daily discharge readings	One reading per well per day. External id unique.	Composite primary key, unique constraint
Groundwater Logger (<code>logger_tmat</code>)	Readings	One reading per logger per timestamp. Alert when the latest depth exceeds 0.4 m.	Composite primary key, <code>peta.v_tmat_alert</code>
Poster Export (<code>ekspor_poster</code>)	None	Progress from 0 to 100. Status is waiting, rendering, done, failed or cancelled. Finishing marks the approved version as printed.	CHECK constraints, <code>tandai_dicetak</code> trigger

Objects that are **facts**, not aggregates, are imported and never edited by planners: hotspots, burned areas, analysis units and contours. They are treated as immutable reference data from the planner's point of view.

Entities and value objects

Kind	Examples	Identity
Entity	KHG, Canal Block, Canal, Pump, Plan Version, Deep Well	Surrogate <code>id</code> (bigint identity) plus a human code such as <code>SK-24</code>
Value object	Location (point geometry), suitability score set, overlay context of a canal block, burn years (<code>smallint[]</code>), peat thickness class	Defined only by its values
Immutable record	Change record, hotspot, reading	Append only

Plan Version lifecycle

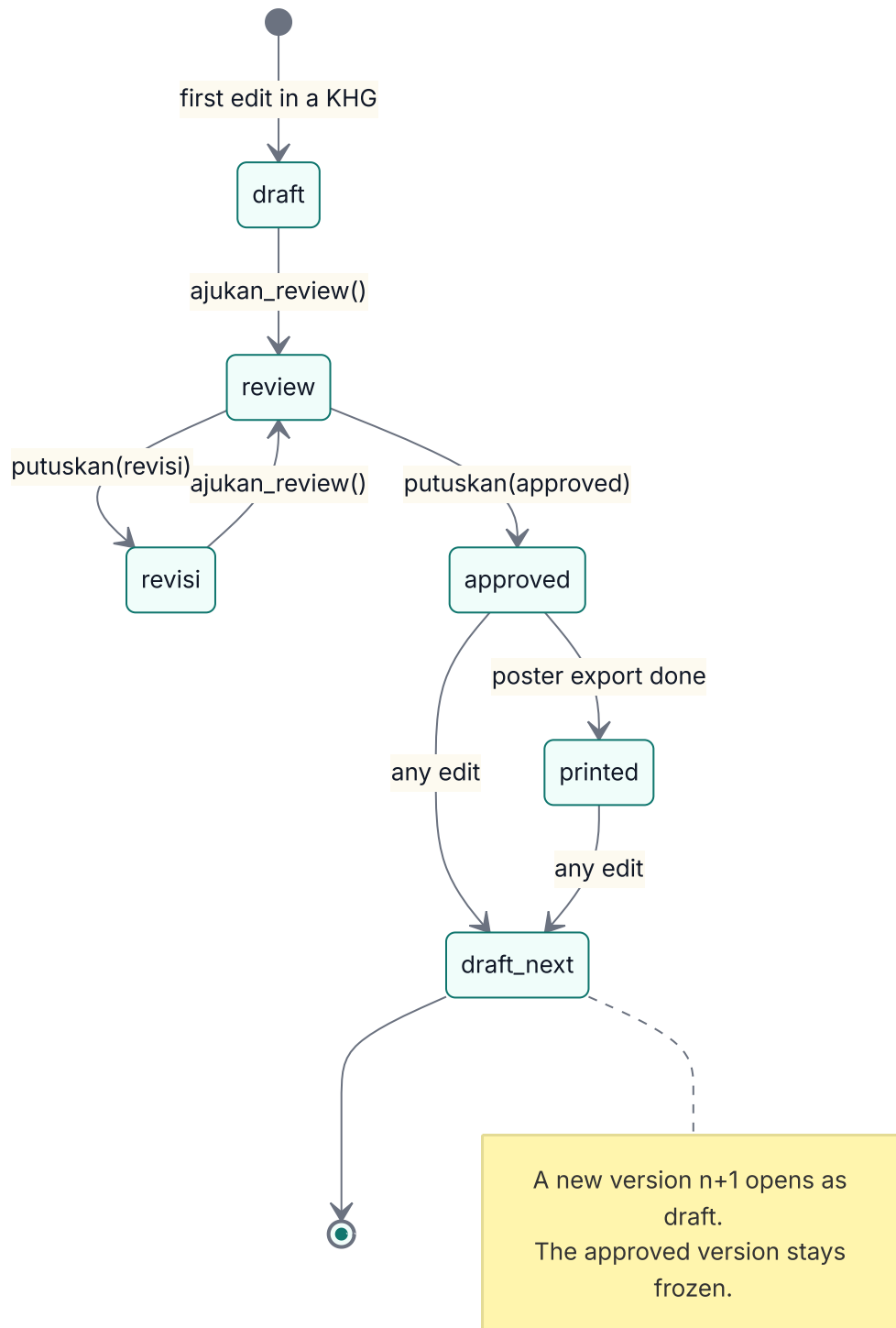


Figure 2. State machine of a plan version.

Domain events

The database raises these events through triggers and functions. They are stored as rows, which makes them auditable and easy to replay.

Event	Raised when	Raised by	Effect
FeatureCreated, FeatureUpdated, FeatureDeleted	A planning feature changes (QGIS, web or API)	<code>z_audit</code> trigger (<code>alur.catat_perubahan</code>)	Change record with old and new values, user and channel, linked to the open version
VersionOpened	The first change after approval, or the first change ever	<code>alur.versi_terbuka()</code>	New draft version n+1
VersionSubmitted	Analyst submits for review	<code>alur.ajukan_review()</code>	Status review and a frozen validation snapshot
VersionApproved, RevisionRequested	Reviewer decides	<code>alur.putuskan()</code>	Status approved or revisi, reviewer note
PosterExportCompleted	Render job finishes	Update of <code>ekspor_poster.status</code>	Triggers VersionPrinted
VersionPrinted	Approved version was exported	<code>tandai_dicetak</code> trigger	Status printed
StatisticsRefreshed	Daily job or manual request	<code>api.refresh_statistik()</code>	KHG statistics recomputed

Unchanged saves (for example QGIS saving a layer with no edits) do **not** raise events. The trigger compares old and new values first.

Domain services

Service	Question it answers	Implementation
Validate plan version	Is this plan ready for review?	<code>alur.validasi_khg(version, max_distance)</code> : pumps near water, pumps outside concessions, canals surveyed, pump capacity filled, valid geometry
Suggest pump locations	Where should the next pump go?	<code>peta.saran_lokasi_pompa(khg, max_distance, min_peat, concession_buffer)</code> : samples active canals and scores distance to river, canal class, peat thickness and burn history
Prioritise KHG	Which KHG need attention first?	<code>api.khg.skor_prioritas</code> : 40 percent burned share, 30 percent deep peat share, 30 percent recent hotspots
Allocate codes	What is the next human-readable code?	<code>alur.isi_kode()</code> trigger: P-n , SK-n , K-n per KHG with an advisory lock
Locate KHG	Which KHG does a new feature belong to?	<code>alur.isi_meta()</code> trigger: spatial lookup when <code>khg_id</code> is empty

Business rules catalogue

No	Rule	Type	Where
R1	Every planning feature belongs to at most one KHG, derived from its location when not given.	Invariant	<code>isi_meta</code> trigger
R2	A KHG has at most one open plan version at any time.	Invariant	Partial unique index on <code>khg_versi</code>
R3	Approved and printed versions never change; new edits open a new version.	Invariant	<code>versi_terbuka()</code>
R4	Every change records who made it, with which tool and when.	Policy	<code>catat_perubahan()</code> as SECURITY DEFINER
R5	A pump should be within 500 m of a river or active canal.	Validation (error)	<code>validasi_khg()</code>
R6	A pump must not be inside a concession.	Validation (error)	<code>validasi_khg()</code>
R7	Every canal must have a survey status before approval.	Validation (error)	<code>validasi_khg()</code>
R8	Every pump should have a capacity.	Validation (warning)	<code>validasi_khg()</code>
R9	Pump suitability scores lie between 0 and 1.	Invariant	CHECK constraints
R10	Groundwater deeper than 0.4 m below surface raises an alert.	Policy	<code>peta.v_tmat_alert</code>
R11	Public users may read everything; only authenticated editors may write.	Policy	Caddy proxy and database roles
R12	Web clients must not overwrite someone else's newer edit.	Policy	Optimistic lock on <code>rev</code>

Mapping to the implementation

Context	Tables	OGC API collections and processes
Reference	ref.*	khg , desa
Infrastructure Planning	peta.kanal , sekat , sekat_konteks , sekat_bangunan , pompa , pintu_air , sungai	sekat , sekat-edit , kanal , pompa , pintu-air , sungai ; process saran-lokasi-pompa
Risk	peta.hotspot , areal_terbakar , ketebalan_gambut , konsesi	hotspot , areal-terbakar , ketebalan-gambut , konsesi
Field Monitoring	peta.logger_tmat , tmat_bacaan , sumur_bor , sumur_debit , posko_karhutla	logger-tmat , sumur-bor , posko-karhutla
Spatial Analysis	analisis.*	unit-nasional , unit-target- 2026 , kontur , kontur-lidar
Plan Review	alur.*	processes validasi-khg , ajukan-review , putusan- review , ringkasan-dashboard
Publication	api.* views	all read collections, tiles, refresh- statistik

Open questions for domain experts

1. **Priority score.** The weights (40, 30 and 30 percent) are provisional. What formula does the directorate use?
2. **Pump suitability.** The weights for water distance, canal class, peat thickness and burn history (35, 25, 20 and 20 percent) are provisional.
3. **KHG codes.** The official codes such as KHG.14.12 are not in any source dataset. Which decree lists them?
4. **Canal status.** Who confirms whether a canal is active, and how often?
5. **Hotspot source.** Should monthly counts use one satellite source only? The current data mixes MODIS-only months with MODIS plus VIIRS months, which biases monthly trends.